BIOCHEMISTRY CUMULATIVE EXAM
DECEMBER 7, 2001

Assume T = 25°C = 298°K
$\Delta G^o = -RT\ln K_{eq} = -RT2.303\log K_{eq}$
pH = p*K* + log [A⁻]/[HA]
R = 1.987 cal/(mol·K)

1. Calculate the equilibrium constant K_{eq} for each of the following reactions assuming first that the equilibrium concentration of each reactant and product are 1 M and then 1 mM. Include units in the constants. Using the equilibrium constants calculated using 1 mM for equilibrium concentrations, determine the equilibrium concentration of products if the reactants are 1 M. (If there are two products assume they are equal in concentration.) 20 points

a) $A \Leftrightarrow B$

b) $A \Leftrightarrow B + C$

c) $2A \Leftrightarrow B$

d) $A + B \Leftrightarrow C + D$

2. The reaction $A + B \Leftrightarrow C + D$ has a $K_{eq} = 10$. If the reaction is started by mixing 10 mM A with 10 mM B, what will the final concentrations of A, B, C, and D be at equilibrium? 15 points

3. The reaction $A + B \Leftrightarrow C + D$ is started by mixing together 4 mM A, 3 mM B, 2 mM C and 1 mM D. After equilibrium is reached, the concentration of B was determined to be 1 mM. What is the value of K_{eq} for the reaction? 15 points

4. Write the Henderson-Hasselbalch equation. 15 points

5. Calculate the hydrogen ion concentration at the pH value: a) 1 b) 3.5 c) 6.5 d) 8.7 e) 9.8. Calculate pH if the hydrogen ion concentration is f) 1×10^{-3} g) 1×10^{-5} h) 1×10^{-7} i) 1×10^{-10}. 15 points

6. a. What volumes of 0.1 M KH_2PO_4, 0.1 M K_2HPO_4, and water must be mixed to make 1 L of 0.1 M phosphate buffer at pH 7.4? The relevant p*K* of phosphate is 6.86. 10 points
b. Suppose you dilute this pH 7.4 buffer so that it is now 10 mM in phosphate. What is the pH now? 10 points